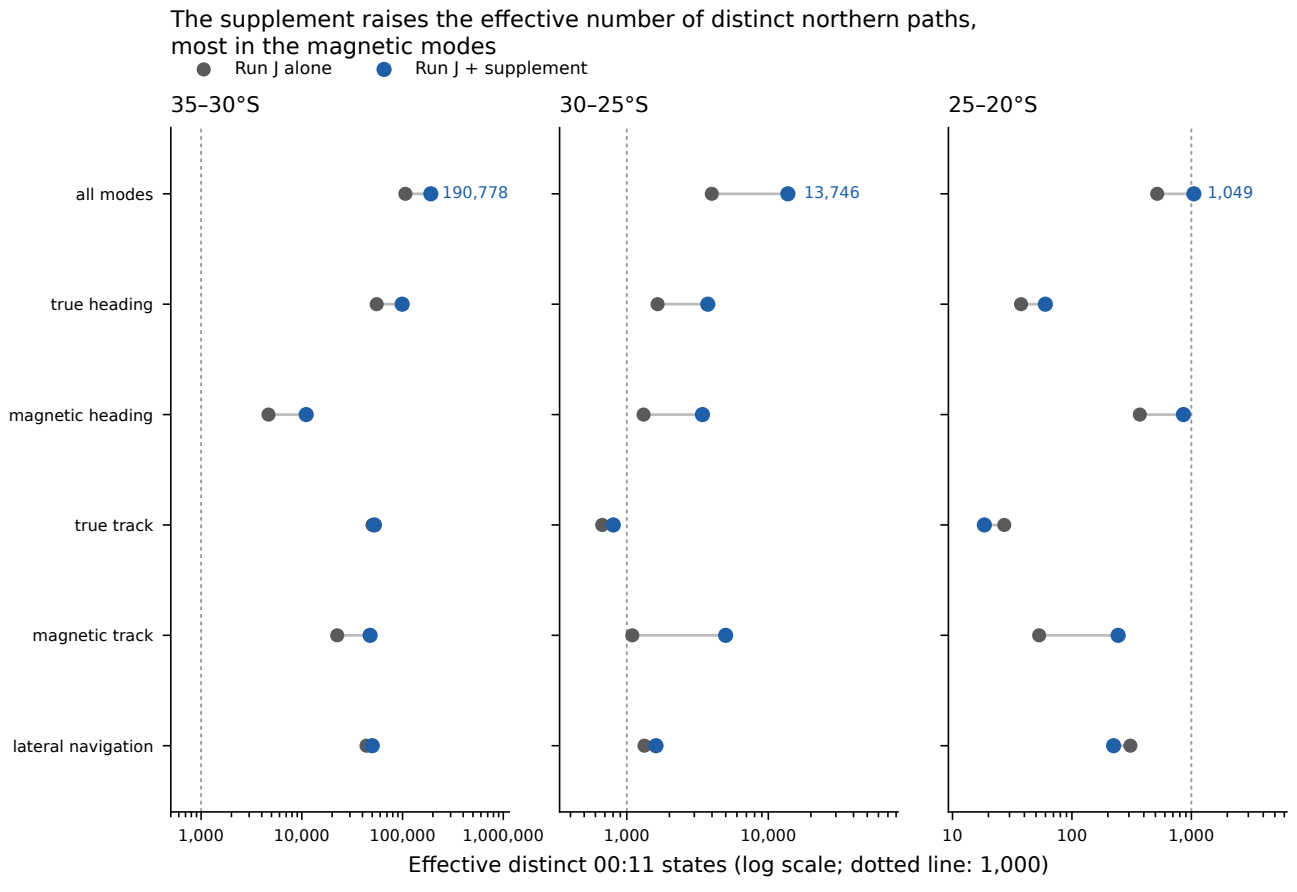

Plain: Each dot gives the effective number of different aircraft states at 00:11 in one latitude band. Copies of one state count once. Grey is run J alone. Blue is run J and the supplement together. The supplement gives about 3 times more in 30 to 25 S and 2 times more in 25 to 20 S. The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: Kish ESS over distinct (seed, state) pairs, state = (latitude, longitude, altitude, BFO bias mean) of the 00:11 hand-off row; posterior weights (rule B for combined, w exp(lc)/24 for run J alone). Bands of 00:11 latitude (about 1 deg north of the impact band). Distinct states production -> combined: 35-30 S 461k -> 825k; 30-25 S 15.4k -> 90.0k; 25-20 S 1,966 -> 13,705. Under D2-D4 the effective count is lower (it depends on the 00:19 likelihood); end of flight measures it on impacts. Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.